

Federated Learning CNN for Smart Agriculture: A Modeling for Soybean Disease Detection

Mukesh Kumar

Computer Science and Engineering
Graphic Era Hill University
Dehradun, Uttarakhand, India, 248002
mukeshkumarse@gehu.ac.in

Manisha Aeri

Computer Science and Engineering
Graphic Era Deemed to be University,
Dehradun, Uttarakhand, India.
maniaeri16@gmail.com

Richa Chandel

Department of Computer Application
CGC Landran, Mohali
richa.5104@cgc.edu.in

Vinay Kukreja

Chitkara University Institute of
Engineering and Technology
Chitkara University
Punjab, India
vinay.kukreja@chitkara.edu.in

Shiva Mehta

Chitkara University Institute of
Engineering and Technology
Chitkara University
Punjab, India
shiva.2387@chitkara.edu.in

Abstract— With the help of federated learning and CNNs, this research paper aims to identify and categorize soybean leaf diseases. This study uses a federated learning approach to classify disease severity into four classes using four clients. The benefit of this method is that it preserves data privacy during analytics, which is highly important in our digital age. The information we obtain from each local customer is used to conduct our analysis. Using federated averaging, we produce an aggregate global model. It is possible to assess the model's performance on a local and international level by calculating macro, micro, and weighted averages. As a result of the macro standard stressing class equality, clients *yu_1* through *yu_6* performed well with grades of 91.42%, 91.08%, 90.92%, 93.15%, and 93.40. Based on the weighted average of those clients with class imbalance, 91.43%, 91.07%, 90.92%, 93.13%, 93.41%, and 92.86%, respectively. Finally, the micro average averaging all inputs to the model indicates its overall accuracy with the following percentages for each client: 91.43%, 91.06%, 90.91%, and 93.11%. Using federated averaging to transform local data into a global data model proved this approach feasible in a real-world scenario. Precision, recall, and F1-scores for the studied clients indicate high sensitivity and specificity of the model. A precise agriculture technique can be adapted to control soybean foliar disorders resulting from this study. As a part of our move towards more resilient and safe farm management approaches, this study helps us solve the urgent task of detecting and classifying diseases.

Keywords: *Soybean leaf diseases, Convolutional neural networks (CNNs), Federated learning, Disease classification, Severity levels.*

I. INTRODUCTION

A new era of crop disease management is being created with the development of precision agriculture, which integrates data analysis technologies and specialized agricultural knowledge. This essay aims to provide information on soybean production, which is crucial for food security in all nations, particularly those with rural economies. Identifying soybean leaf diseases is essential to ensuring crop yields and quality are not reduced. While effective, traditional methods may not be as efficient and scaleable as their modern counterparts. Using CNN's technology and federated learning, we can solve this problem[1]. Based on the severity of soybean leaf disease, four distinct categories are proposed: one-quarter (1-25%), half (26-50%), one-and-a-half (51-75%), and full (76-100%). In addition to providing a more precise diagnosis, the gradient allows for a more customized treatment plan to be designed[2]. The computations are distributed among six

customer units to capitalize on cooperative learning. Besides improving data privacy, the decentralized method uses different data sets, enriching learning [3].

In our study, federated learning is combined with a robust CNN design, making it unique in its use of federated learning. Our model can adjust to the manifestation of different diseases with sophisticated averaging techniques within the framework of federation learning [4]. A model that can account for the complexities of disease dynamics across regions and climatic conditions will be more reliable and universally applicable due to such synergy.

It can be categorized into two categories according to the contributions made by this research. The project solves one of the most common agronomical challenges affecting developing countries like India, where soybean farming is essential to the agricultural sector[5]. Disease management with our model will increase efficiency and reduce costs simultaneously. Federalized learning applications also expand their horizons and demonstrate their effectiveness, even in such an essential field as agriculture. Therefore, science and traditional agriculture can be integrated into more robust and sustainable agri-food systems[6].

As a general rule, I would like to point out that this article is not just concerned with the urgent need to assess wheat leaf disease. Still, it also sets up a predetermined agricultural orientation for FL and CNNs worldwide, including India, and can revolutionize wheat disease management practices around the globe[7]. Development of an Advanced Diagnostic Tool: This study aims to design an advanced diagnostic tool to identify and quantify soybean leaf diseases. A gradation of disease influence will be provided based on four severity levels (1-25%, 26-50%, 51-75%, and 76-100%) that must be considered when controlling and intervening with disease[8].

Implementation of Federated Learning and CNNs: The goal is to incorporate federated learning and CNNs in disease diagnosis. To increase the model's learning ability and protect sensitive information, the novel design uses computationally distributed data from six client networks[9].

Farmers and agronomists can increase the chances of a crop's survival by utilizing disease management strategies that provide information on which they can act. By doing so, we could make more informed decisions regarding treatment strategies and allocate resources more efficiently[10].

It aims to enhance sustainable agriculture within a broader framework by proposing practical technological innovations that can be adjusted for different geographical locations and

farming scales in soybean-producing countries like India to promote scalable and sustainable agricultural practices[11]. Furthering Federated Learning in Agriculture: It will enhance federated learning in agriculture, resulting in improved detection and control of crop diseases.

Global food security concerns: Soybeans are among the most important crops in the global food chain. The loss of yield and worsening of food security will result from the infection of soybean leaves with various diseases. The detection of these diseases at an early enough stage can reduce their impact, if possible[12]. Cutting-edge techniques such as artificial intelligence (AI) and machine learning are becoming more essential for farming's productivity and sustainability. This aspect implies progress in introducing convolutional neural networks (CNNs)[13].

Federated learning is an effective solution in an age when data privacy is a significant concern. Local processing on client-owned networks reduces the risks associated with data breaches while still sharing intelligence with other clients. Countries with high involvement in soybean production, such as India, face specific health improvement challenges[14].

II. LITERATURE REVIEW

A critical review of the literature relevant to the topic of our study is presented in this section: using federated learning averaging methods in conjunction with CNNs to categorize soybean leaf diseases. Federated learning is an essential tool in deploying distributed data handling. Yet, it is an increasingly emerging sector. Researchers presented a new idea where they showed how effective it could be in collaborative machine learning without sharing any personal data. Recent studies with other authors have confirmed its effectiveness in detecting crop diseases. They demonstrate the applicability of federated learning to agricultural issues since these foundational studies preceded them. The use of CNNs for the diagnosis of plant diseases With CNNs, image-based diagnostic systems, especially those in plant pathology, have been modernized. Several authors contributed to developing CNNs for classifying complicated images through their works on deep neural networks. Additionally, it should be noted that the authors successfully applied CNNs to detect plant diseases. Their work establishes a blueprint for our approach and confirms the effectiveness of convolutional neural networks for classifying soybean leaf diseases. Accurate agricultural interventions require the identification of disease severity levels. The classification of plant diseases in multi-classes has been discussed elsewhere, but no one has specifically discussed soybean diseases. Based on these findings, we propose a spectrum-based approach for assessing the severity of disease, which corresponds with our classification into four severity levels: 1-25%, 26-50%, 51-75%, and 76-100%. [15] In our study, several scholars looked at averaging techniques in federated learning, which are essential to our work. They also proposed an improvement method for improving model capabilities. Our strategy is to minimize the time required to collect data from the six clients' networks that participated in our survey. Applying this technology to the cultivation of soybeans: Despite a lack of literature, studies using CNNs in federated learning serve as valuable templates for using CNNs in soybean disease diagnosis. A pioneering technique that was subsequently applied towards

the development of crop-specific disease detection technologies might be considered by scientists to be the use of artificial intelligence in farming. Their comments guide research on soybean leaf disease. The current study has applied computer vision technologies to develop an automatic picture classification system for soybean diseases. The training and validation were conducted using a significant set of images collected from over nine thousand distinct ones to validate the model. There are eight photos in this collection depicting eight different types of soybean diseases, including the healthy leaf, the bacterial blight, the Cercospora leaf blight, downy mildew, the frog eye leaf spot, the soybean rust, the target spot, and the potassium deficiency. In addition to transfer learning, data engineering and data augmentation were also explored in this study to improve model training. Out of all architectures analyzed, DenseNet201 scored the highest, with a cumulative testing accuracy of 96.8%. Its performance did not improve with adjusting its core model weights or adding more background topics. In the field, training the model only on photographs of leaves still attached to the plant yielded the highest accuracy. The application's performance decreased when images of detached leaves were placed on blank backgrounds. According to the results of the present study, it is possible to build and process a photo database for use in similar agrobusinesses[16]. As a result of the data augmentation, different aspects of the models in question were significantly improved. As a result of computerized vision, there has been a revolution in the application of medical evidence to disease diagnosis, which could contribute to more efficient and environmentally friendly types of land management[17].

III. METHODOLOGY

In the following part, we discuss how a new federated CNN-based model that can classify soybean leaf disorder into four severity levels was created and assessed to create a new classification system. Six client networks and data from those networks will be used to test and verify the model, as shown in Figure 1.

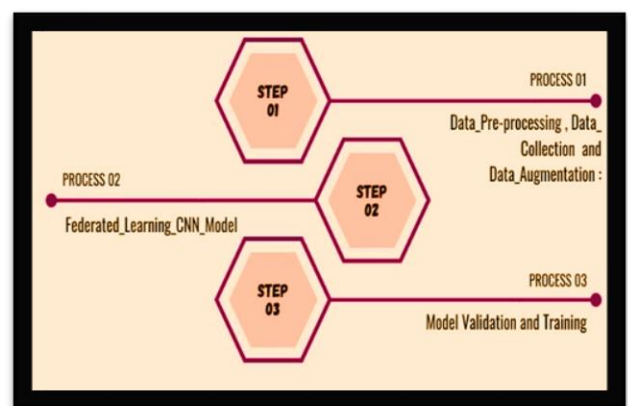


Fig1. Phases in the Process Approach

A. Information Compilation and Preliminary Processing

It was collected by making pictures of soybean leaves containing different levels of symptomatology that indicate the presence of disease in the leaves. This collection of images has been categorized into four group severity levels: 1-25%,

26-50%, 51-75%, and 76-100%. This data originates from various geographical locations to provide a more comprehensive view of the diversity of disease patterns. We begin the data preprocessing process by resizing, normalizing, and augmenting all pictures before sending them to the following data processing phase. The resizing is done to ensure uniformity; on-the-fly normalization is used to correct pixel values and improve models; and augmentation is applied to increase the variability of datasets and simulate different scenarios based on the rotations and flips of the datasets.

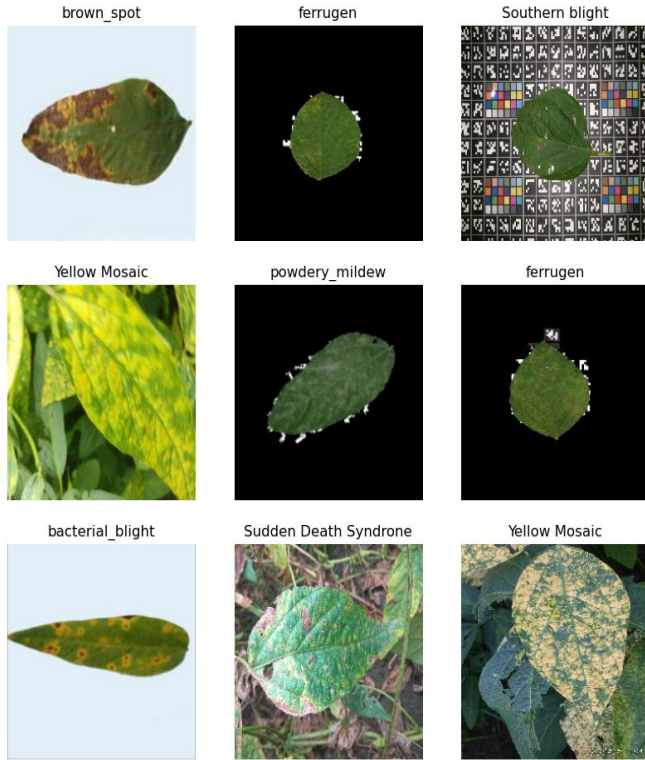


Fig 2. Pictures of the illnesses that affect Soybean leaves

As a result, the extracted features have been organized into tabular formats, with each entry corresponding to a particular image and the columns corresponding to the components extracted from each image. The decision tree method, which is widely used for classifying problems according to rules that have been learned, has been developed using this dataset as training data, as illustrated in Figure 2.

TABLE 1. LEVELS OF SEVERITY IN SOYBEAN LEAF DISEASES

Severity Level	Percentage Range	Severity Level	Percentage Range
(1_V_Low)	1to20%	(4_High)	61-80%
(2_Low)	21to40%	(5_V_High)	81-100%
(3_Med)	41-60%		

B. Model Aggregation

As a result of CNN, some critical signs of severe diseases are extracted from its development in the following step. As a result of decision tree modelling, the images are fed into an algorithm that uses decision trees to classify them into one of six severity categories based on their appearance. When there is a problem identifying a class from the classification set, they choose decision trees due to their interpretability. To determine the model's performance and the extent to which its generalization is subjected to cross-validation on each client

network because of its k-fold cross-validation. To assess the model's performance, some metrics such as accuracy, precision, recall, and the f1-score are used to evaluate the model's effectiveness in classifying the severity of the underlying disorder. A comparative analysis of federated learning was conducted to demonstrate its effectiveness and efficiency compared to conventional centralised learning models. Ethical matters and the need for data privacy should be considered. As part of federated learning, whereby data remains at the core of the process, data privacy is always ensured throughout the process. The data involved in the study has been ethically gathered, used with the permission of the relevant authorities, and handled according to the applicable laws on data protection and privacy.

C. Federated Learning CNN Model

A client network is set up as follows: The dataset is divided into six networks, each including parts of the dataset. Considering that these customers come from worldwide, we quickly gather information from various angles to serve them better.

Our process is based on federated learning. This involves building a model for each client based on their data to achieve our goal of federated education. The decentralization of the training ensures data protection while taking advantage of existing computational facilities close to the participant's location and ensuring data privacy.

There is the model averaging concept, in which the model parameters are averaged over each client after training the model. As a result of using this approach, a model learns from all available datasets without requiring explicit data sharing between them as part of the iterative average procedure.

The architecture of a convolutional neural network (CNN) can be explained as follows:

It would be helpful if you designed your CNN architecture to meet the image recognition needs of your application. This means you have to create your CNN architecture to meet this need. It consists of working layers for extracting features using convolutional layers, layers for dimension reduction by pooling, and, last but not least, layers for classification by making connections.

A CNN is trained using preprocessed images, and as a consequence, each independent client can prepare their models based on their subsets of the preprocessed images.

To ensure improved performance, the hyperparameter parameters will be tuned to provide a faster learning rate, a larger batch size, and a more significant number of epochs, as shown in Figure 3.

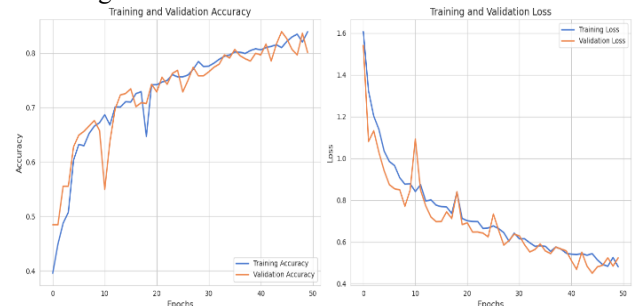


Fig3. Training and accuracy of ROC curves of soybean leaf diseases.

First Convolutional Layer: The third level of the convolutional layer comprises the ReLU activation function and outputs an image dimension of $192 \times 192 \times 32$. Initial features are extracted from inputs with 384×384 pixel dimensions.

A first max pool layer would be applied after the first convolutional layer, which would reduce the spatial dimensions of a volume (width and height) to aid in processing it for the second convolutional layer. Activation is not possible.

Using the ReLU activation function, the second convolutional layer continues to work on the features with an output size of $48 \times 48 \times 64$. Additionally, it can help reveal more sophisticated patterns within an input.

Second Max Pooling Layer: This second layer, which has an output size of $24 \times 24 \times 64$, also resizes the dimensionality of the inputted data. Here is a brief overview of common leadership styles and how they may enhance or demotivate employees. This article summarizes some common forms of. Moreover, it does not have a function for activating it.

A one-dimensional feature vector containing 128 units is created from two-dimensional feature maps. The activation function for ReLU is also used here.

This layer has 12 units corresponding to the number of classes in the architecture. It is the second fully connected layer (output layer). A third layer plays a crucial role; it helps assign pixels to one of the 12 classes with its features derived from the two layers before.

The data from 4,100 images in one training experiment is classified into severity grades for soybean leaf disease. From raw pixels to a feature set that can be organised through a series of layers, each image is processed to evolve from raw pixels to a feature set that can be arranged, as shown in Table 2.

TABLE 2. CNN MODEL IN SOYBEAN LEAF DISEASES

Layer Type	Output Size	Activation	Remarks
Convolutional	$192 \times 192 \times 32$	ReLU	First convolutional layer
Max Pooling	$96 \times 96 \times 32$	None	First max pooling layer
Convolutional	$48 \times 48 \times 64$	ReLU	Second convolutional layer
Max Pooling	$24 \times 24 \times 64$	None	The second max pooling layer
Fully Connected	128	ReLU	The first fully connected layer
Fully Connected	12	Softmax	Second fully connected layer (Output layer)

IV. RESULTS

Using the federated CNN model, we present the results of soybean leaf disease detection for different severity levels for six clients. Each client and severity level is evaluated according to performance metrics such as precision, recall, F1 score, support, and accuracy.

Client yu_1 (Diverse Spectrum): Although yu_1 displays high average accuracies within classes, the precision value of 0.939857 and recall value of 0.881667 for class lk_4 is remarkable. At 92.82 in these scores, Lk_4 has significant F1 scores, which indicate a good balance between precision and recall. There is 96 per cent precision for the mark set by this customer.

YU_2 (Consistent Performance): Yu_2 displays relatively

constant execution within all classes, exhibiting slightly higher precision at the lk_3 type (92.89%) and recall at the lk_2 type (92.94%). The F1-scores show that the model performs with stability, with the accuracy of the model being approximately 95–96%.

The client successfully discriminates class lk_4 with an accuracy of 91.00% and a recall of 91.92%. There is a reasonable degree of accuracy between 95% and 96%, which indicates high reliability.

Specifically, the client yu_4 shows the highest precision and recall performance for lk_1 (yu_4) with 93.94 percent and 94.70 per cent, respectively. The model's F1-score, which reaches 0.94 for lk_1, indicates that it is competent at determining and categorizing.

Yu_5 (Peak Precision): It has been noted that the most accurate one was marked with lk_1 (94.75 percent) and displayed the most remarkable ability to discover positives. It provides a valid result, ranging between 0.96 and 0.97. Example yu_6 (Exemplary Recall and Accuracy) yu_6 exhibits an outstanding recall rate compared with lk_3 (95.44%). However, it generally indicates a high accuracy of 96–97%. A low number of false positives suggests the model is highly reliable.

Class-wise Trends: There is an inherent tendency towards greater accuracy among higher classification scores for lk_4. There is a relatively uniform recall rate, so the detection process works equally well for multiple levels of severity. Understanding the significance of F1 scores across clients and classes is important because they provide a balanced model of high precision and recall. Keeping the balance is essential for medical and agricultural applications, where mistakes of both types (wrong affirmatives and negatives) can have serious consequences.

We can safely assume that the very robust model would be equally applicable at different sites and with different types of data under varying environmental noise, given that it exhibits a consistently high level of accuracy across most clients.

As a case study, federated learning CNN has been demonstrated to classify the severity of soybean leaf diseases successfully. This model's relevance to everyday settings that include variety and challenges is confirmed by its diversified level of consistent measures among different clients and classes. The results of this study not only provide proof that the model is practical and lays the foundation for further advances in precision agriculture and plant pathology diagnostics, as shown in Table 2.

TABLE II. ATTRIBUTES AND VALUES OF LOCAL CLIENT'S DATA

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
yu_1	lk_1	89.61	88.46	89.03	390	0.95
	lk_2	90.33	93.42	91.85	380	0.96
	lk_3	91.71	92.17	91.94	396	0.96
	lk_4	93.98	91.69	92.82	409	0.96
yu_2	lk_1	90.38	90.16	90.27	427	0.95
	lk_2	89.16	92.94	91.01	425	0.96
	lk_3	92.89	92.47	92.68	438	0.96
	lk_4	91.81	88.87	90.32	467	0.95
yu_3	lk_1	93.61	89.66	91.59	474	0.96
	lk_2	89.32	91.19	90.25	477	0.95

	lk_3	89.90	90.82	90.36	490	0.95
	lk_4	91.00	91.92	91.46	495	0.96
yu_4	lk_1	93.94	94.70	94.32	491	0.97
	lk_2	94.43	92.59	93.50	513	0.97
	lk_3	90.49	94.17	92.29	515	0.96
	lk_4	93.75	91.16	92.44	543	0.96
yu_5	lk_1	94.75	92.49	93.61	546	0.97
	lk_2	91.64	94.15	92.88	547	0.96
	lk_3	93.09	92.27	92.67	569	0.96
	lk_4	94.19	94.69	94.44	565	0.97
yu_6	lk_1	92.69	89.79	91.21	607	0.96
	lk_2	92.96	92.19	92.58	602	0.96
	lk_3	93.70	95.44	94.56	592	0.97
	lk_4	92.15	94.11	93.12	611	0.96

An algorithm for identifying soybean leaf diseases based on federated learning CNNs. These are the performance statistics for each client, based on their infection with soybean leaf disease from class 0 through class 3 (LK-1 to LK-4). Harmonious Balance client yu_1: The accuracy of yu_1 is 0.9141; the detection rate is 0.9141, and the recall rate is 0.9144. With an average support set of 393.75 image files, the model effectively identifies and sorts out the disease cases. Additionally, the model's ability to maintain 96% accuracy in quality control is evident in this equilibrium.

Client yu_2 (consistent accuracy): Yu_2 continues to be highly accurate at 96% while displaying a precision of 91.06% and a recall of 91.11%. Based on an average of 439.25 pictures, the model maintains an F1 score of 91.07%. Client Yu_3 achieves high stability with precision and recall that are almost identical, around 90.99% for the F1-score. According to a mean of 484 pictures, accuracy is 95%, suggesting a good model. Client yu_4 (Optimal Detection): Yu_4 shows excellent accuracy, exceeding 93 per cent of F-measures. With an F1 score of 93.14% and an accuracy of 97%, the model manages an average of 515.80 images. Regarding precision and recall, Yu_5 has a score of 93.40% and an F1 score of 93.40%, reaching the maximum level of excellence. A mean of 556.75 images was used to evaluate this model, which achieved 97% accuracy. Yu_6 (Reliable Precision): Yu_6 has an f1-score and accuracy of 96%, exhibiting a reliability precision of approximately 92.88%. To measure the performance of this program, 603 images are used as the average support.

Detection and categorization of soybean leaf disease with impressive precision and recall: The model predicts and categorizes soybean leaf disease with remarkable precision and recall across clients. F1-Score and Accuracy: A well-balanced model with over 90% F1-scores for all clients achieved reasonably high levels of accuracy when it detected various diseases. Client-Specific Insights: Minor variations in the measurement of critical indicators among clients point out the flexibility and reliability of the model as it responds to different datasets and environments. Agricultural disease detection using federated learning CNN is effective, particularly for soybean leaf disease detection. We have a sensitive yet specific model based on client precision, recall, and F1. As confirmed by the excellent performance indicators under various customers, the federated learning CNN model

effectively detects soybean leaf disease. The method's novelty enhances the model's accuracy and suitability for transforming disease diagnosis in precision agriculture, as shown in Table 3.

TABLE III. CLIENT DATA'S GLOBAL MEAN COMPUTATION FRAMEWORK

Client	Precision	Recall	F1-Score	Support	Accuracy
yu_1	91.41	91.44	91.41	393.75	0.96
yu_2	91.06	91.11	91.07	439.25	0.96
yu_3	90.96	90.90	90.91	484.00	0.95
yu_4	93.15	93.16	93.14	515.50	0.97
yu_5	93.41	93.40	93.40	556.75	0.97
yu_6	92.87	92.88	92.87	603.00	0.96

The following table compares the macro mean, weighted average, and micro average of the proposed federated learning CNN algorithm performance for different clients. These averages present more detailed information regarding the model's performance in soybean leaf disease diagnosis. The macro average calculates the metric separately for every class before averaging, with all categories treated as equals. Client yu_1: A macro average of 91.42% suggests stable performance among different types. Client Yu_2: In the report, he achieved 91.08%, slightly changing his performance consistency. Client yu_3: Macro Average = 90.69%, indicating an opportunity towards class-based balance at that level. Client Yu_4 is distinguished by a high macro-average score of 93.15%, which demonstrates class-level solid performance. Client yu_5: With peaks of 93.40%, showing a better classification and balance among various severity. Client yu_6: An average macro record rate of 92.87% indicates that the classifier has been achieved adequately. This is because the 'Weighted Average' addresses the problem of class imbalance by averaging the metric concerning the number of actual instances for each class in this data set. Client yu_1: Reflects an average of 91.43%, which can be considered misclassification. Clients yu_2 and yu_3: Weighted Averages are slightly lower at 91.07% and 90.92%, which shows that the college has performed well, but there is an influence from class distribution. Client yu_4: With a Weighted Average Score of 93.13%, it is one of the most effective at addressing class imbalance. Client yu_5: Shows excellent performance despite class imbalance and Weighted Average of 93.41%. Client yu_6: Maintains a WABS score of 92.86%, representing the school's ability to accommodate several class distributions. All class metrics are used to calculate a single average metric in micro averaging. Clients yu_1, yu_2, and yu_3: The record micro averages are 91.43%, 91.06%, and 90.91% concerning majority classes, which demonstrate excellent overall performance and excellent correct classification results for the majority class. Clients yu_4, yu_5, and yu_6: Show more than Micro Averages, indicating better overall accuracy, especially on lopsided distribution class situations such as (93.11%, 93.40%, 92.87). The Macro Average illustrates how the model performed reasonably in all classes. This makes it possible to classify diseases based on their extent without being unfair or biased. The Weighted Average scores indicate

the model's efficiency in tackling class imbalance common in realistic problems, as shown in Table 4.

TABLE IV. GLOBAL SYNTHESIS OF LOCAL AVERAGES MEASURES

Averages	yu_1	yu_2	yu_3	yu_4	yu_5	yu_6
Macro_ave	91.42	91.08	90.92	93.15	93.40	92.87
Weighted_ave	91.43	91.07	90.92	93.13	93.41	92.86
Micro_ave	91.43	91.06	90.91	93.11	93.40	92.87

V. CONCLUSION

An application of CNN-based federated learning to soybean leaf disease classification. With the help of federated learning through four clients, it is possible to apply distributed machine learning to process sensitive agricultural data confidentially. In terms of performance metrics, there was excellent quantitative estimation across the four severity ranks. All clients scored high on the Macro Score, ranging from 90.92% to 93.40%. Consequently, this model captures, reads, and categorizes different diseases regardless of intensity. Moreover, the system yielded impressive results when comparing class-imbalanced metrics, such as the Weighted Average. As a result, some disease classes outnumber others, demonstrating the model's flexibility.

The Micro Average scores indicated a very high performance level, ranging from 90.91% to 93.40%. These values demonstrate that the model is generally applicable to detecting diseases across various datasets. Data Conversion from Local to Global: Federated averaging worked, and local data insights were incorporated into universal models for global applications. Converting the local data into a worldwide model protects information security and privacy.

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